Who sets the temperature? Polarising rhetoric, voter engagement and the feedback loop between parliament and social media in Sweden and Finland

Research plan · Asad Ikram · PhD in Political Science (PA2026/2879), project "A Divisive Spiral?"

1. Aim and research questions

The project asks whether affective polarisation grows from a spiral in which politicians use polarising rhetoric, voters reward it with engagement, platforms amplify what is engaging, and politicians learn from the signal. My dissertation would study the elite side of that loop: whether politicians are rewarded for polarising communication on social media, and whether the reward changes how they speak afterwards, online and in parliament. Because this comparison crosses arenas, languages and modes, it rests on a measurement question that has to be answered first.

- **RQ1 (measurement).** Can large language models measure polarising communication (out-party hostility, moralised language and emotional appeals) equivalently in parliamentary speech and in social media posts, in Swedish and Finnish, and in text and transcribed audio?
- **RQ2 (reward).** Do politicians' more polarising posts receive more engagement, and does the reward differ by party, platform and electoral period?
- **RQ3 (feedback).** When a politician is rewarded for polarising posts, does their later rhetoric become more polarising, on social media and in parliament?

2. Background and contribution

Affective polarisation, dislike and distrust of political opponents, has risen in many democracies (Iyengar et al. 2019). Elites are a plausible driver: Bäck et al. (2023) show in a survey experiment with about 1,300 Swedish voters that a factual message carrying an in- or out-party source cue raises affective polarisation, especially among strong partisans. Social media can strengthen that channel. Posts about the political out-group draw more engagement than other posts (Rathje et al. 2021), users who are rewarded for expressing outrage express more of it later (Brady et al. 2021), and ranking algorithms can amplify some political content more than others (Huszár et al. 2022). Together these describe the spiral, but they are rarely tested on the same politicians across arenas, and almost never outside the United States and English. Sweden and Finland add a further test: in multiparty systems voters can dislike several parties to different degrees (Wagner 2021), so polarising rhetoric has to be measured together with its target, not only its tone.

A cross-arena test also needs a measure that means the same thing in each arena. Gaynor et al. (2026) find that legislators' ideological positions differ between votes, floor speeches and tweets; whether that is strategy or an artefact of measuring different kinds of text with one instrument cannot be settled without testing the instrument. Sen et al. (2021) call this platform-affordance error. In a methods paper with Dr Ella Haig (Ikram and Haig 2026), I treat each platform as a survey mode and show in a simulation that pooling machine-labelled sentiment across platforms at face value gives the wrong sign of the aggregate on nearly a quarter of days. LLMs now label political text at scale and in many languages (Gilardi et al. 2023; Rathje et al. 2024), but their validity varies across languages and genres. A parliamentary speech and a Facebook post differ in length, register and audience. If a model reads hostility more readily in one, then a finding that politicians are "more hostile on social media than in parliament" may describe the model rather than the politicians.

The dissertation would contribute (a) a validated, multilingual and cross-arena measure of polarising elite communication that the project's other studies can use, (b) evidence on whether engagement rewards polarising rhetoric in two multiparty systems, and (c) a test of whether that reward feeds back into later rhetoric, including in parliament.

Expectations. H1: a politician's posts expressing out-party hostility receive more engagement than their comparable non-hostile posts, and more than posts that are only emotional. H2: the reward is larger during election campaigns and on platforms whose ranking weights engagement most heavily. H3: rewarded hostility predicts more hostility in a politician's later posts, with a weaker spillover into parliament, where the norms of debate restrain it. H0 for RQ1: LLM labels of hostility are not invariant across arenas without correction, so uncorrected cross-arena comparisons overstate the online–parliament gap.

3. Data

Parliamentary speech comes from the open data of the Riksdag and the Eduskunta: plenary transcripts and, where available, audio, covering at least the Swedish elections of 2018, 2022 and 2026 and the Finnish elections of 2019 and 2023. Social media posts by members of both parliaments and their parties (Facebook, Instagram, X and, where accounts exist, TikTok) would be collected with engagement counts through official research access, such as the Meta Content Library and data access for vetted researchers under Article 40 of the EU Digital Services Act. Audio is transcribed with a multilingual speech model and checked on a sample. Only public communication by public office holders is collected, under the project's ethics approval, with data minimisation and no processing of private users' identities. Collecting and maintaining such corpora reliably has been my professional work for eight years.

4. Measurement (RQ1)

Starting from the project's concepts and existing work on emotion in parliamentary debate (Rheault et al. 2016), I would build a codebook for out-party hostility, moralised language and emotional appeals, with examples in Swedish and Finnish, coding the target party of each hostile statement as well as its intensity. Two LLMs label every unit directly in the original language, and their agreement serves as a convergence check. Native-speaker coders label a stratified gold sample for each cell of arena by language by mode. I would then apply the protocol from my paper: a generalizability study to show how much score variance comes from arena, language, model and prompt; tests of measurement invariance and differential item functioning to find where a label behaves differently, for example whether a heated but procedural exchange in the chamber is over-coded as hostile; and score linking where the scales differ. The output is a statement of which comparisons are licensed, and a corrected measure where they are not. This is the first paper.

5. Analysis (RQ2 and RQ3)

Reward. At the post level I would model engagement as a function of measured rhetoric, with politician fixed effects and controls for topic, format, timing and audience size, comparing platforms and campaign with non-campaign periods. Engagement is not exposure, so I would use reach measures where platforms provide them and treat them as an outcome of algorithmic amplification; the causal role of algorithms is tested in the project's experiments. **Feedback.** Using a politician-by-week panel, I would test whether unusually high engagement on a politician's polarising posts predicts more polarising rhetoric in the following weeks, comparing each politician with themselves (following the design of Brady et al. 2021), and whether it spills over into their parliamentary speeches. Party leadership changes and election campaigns provide natural comparison points. Real examples of polarising and non-polarising messages from the corpus can also serve as realistic stimuli for the project's voter experiments.

6. Feasibility and timeline

Year 1: doctoral courses (political behaviour, causal inference), codebook, data collection, and Swedish language courses. **Year 2:** gold-standard coding and the invariance study (paper 1). **Year 3:** reward and feedback analysis (paper 2), and a contribution to the project's experiments. **Year 4:** spillover into parliament (paper 3) and the compilation thesis. The main risks are platform data access and the reliability of Finnish labels. Parliamentary data are open and form a core that does not depend on platform access, and the native-speaker gold samples are built so that weak cells show up early.

References

Bäck, H., Carroll, R., Renström, E. and Ryan, A. (2023). Elite communication and affective polarization among voters. *Electoral Studies* 84: 102639.

Brady, W. J., McLoughlin, K., Doan, T. N. and Crockett, M. J. (2021). How social learning amplifies moral outrage expression in online social networks. *Science Advances* 7(33).

Gaynor, S., Miler, K., Goel, P., Hoyle, A. and Resnik, P. (2026). Express yourself (ideologically): legislators' ideal points across audiences. *Journal of Politics* 88(4).

Gilardi, F., Alizadeh, M. and Kubli, M. (2023). ChatGPT outperforms crowd workers for text-annotation tasks. *PNAS* 120(30).

Huszár, F. et al. (2022). Algorithmic amplification of politics on Twitter. *PNAS* 119(1).

Ikram, A. and Haig, E. (2026). Is the platform part of the measurement? A protocol and simulation study for cross-platform equivalence of machine-labelled policy sentiment. arXiv:2610.16757.

Iyengar, S., Lelkes, Y., Levendusky, M., Malhotra, N. and Westwood, S. J. (2019). The origins and consequences of affective polarization in the United States. *Annual Review of Political Science* 22: 129–146.

Rathje, S., Van Bavel, J. J. and van der Linden, S. (2021). Out-group animosity drives engagement on social media. *PNAS* 118(26).

Rathje, S. et al. (2024). GPT is an effective tool for multilingual psychological text analysis. *PNAS* 121(34).

Rheault, L., Beelen, K., Cochrane, C. and Hirst, G. (2016). Measuring emotion in parliamentary debates with automated textual analysis. *PLoS ONE* 11(12).

Sen, I., Flöck, F., Weller, K., Weiß, B. and Wagner, C. (2021). A total error framework for digital traces of human behavior on the web. *Public Opinion Quarterly* 85(S1): 399–422.

Wagner, M. (2021). Affective polarization in multiparty systems. *Electoral Studies* 69: 102199.